

ZARYAB MUSTAFA

AI Engineer · Software Engineer · ML Engineer

Pakistan | +92 318 0997343 | zaryabmustafa27602@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI & Software Engineer focused on LLM applications, RAG systems, and scalable backend services. Experienced with FastAPI, LangChain, vector search, and ML workflows, delivering AI systems with <3s response times, >85% model accuracy, and 1,000+ concurrent-user capacity. NLP researcher with an ACL-submitted paper on efficient LLM fine-tuning.

TECHNICAL SKILLS

Languages	Python, JavaScript, SQL, HTML, CSS
Frameworks	FastAPI, Next.js, Node.js, LangChain, SQLAlchemy, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
AI / ML & NLP	OpenAI API, Whisper (ASR), Hugging Face Transformers, LLM Fine-tuning, RAG Pipelines, Vector Search, Argument Mining, Embeddings, Prompt Engineering
Databases & Cache	PostgreSQL, Redis, FAISS, pgvector
DevOps & Tools	Docker, Git, GitHub, Celery, RabbitMQ, gRPC, Nginx, Webhooks
Architecture	REST APIs, WebSockets, WebRTC, JWT, OAuth2, RBAC, Async Processing, Microservices, Circuit Breaker
Deployment	Vercel, Railway, Docker Compose

FEATURED PROJECTS

Oratoria AI · [Final Year Project](#) · [Deployed — Live](#)

Stack: Python · FastAPI · Next.js 16 · OpenAI Whisper · Hugging Face · MediaPipe · OpenCV · Qwen/Gemini/Grok · PostgreSQL · Redis · Docker · gRPC · WebSockets

- Co-developed a multimodal AI coaching platform combining NLP, Whisper (>85% accuracy), and Computer Vision for real-time text, audio, and video feedback.
- Engineered Python/FastAPI microservices backend with gRPC inter-service communication between a locally hosted Whisper transcription service and a multi-provider LLM evaluation service (Qwen/Gemini/Grok via provider router), achieving <3s end-to-end AI analysis response time
- Built multi-dimensional scoring engine across 4 evaluation dimensions — Matter/Content (40%), Manner/Delivery (30%), Method/Structure (20%), Clash/Engagement (10%) — generating adjudicator-style NLP feedback supporting Asian Parliamentary and British Parliamentary debate formats
- Implemented secure authentication, RBAC, WebRTC/WebSockets, and a PostgreSQL/Redis backend designed for 1,000+ concurrent users.
- Built a gamified Next.js frontend with Docker deployment and GDPR/FERPA-compliant data handling.

AuthForge · [Public](#) — [GitHub](#)

Stack: Python · FastAPI · SQLAlchemy · PostgreSQL · Redis · Celery · JWT · OAuth2 · Docker · Nginx · pytest

- Architected a production-grade authentication platform comparable to Auth0/Clerk, supporting JWT stateless auth, refresh token rotation, multi-provider OAuth2 (Google, GitHub), and device-aware session management with Redis-backed token blacklisting
- Engineered secure refresh-token rotation with Redis locks and replay-attack mitigation.
- Implemented RBAC, rate limiting, OTP verification, brute-force protection, and SQL injection safeguards.
- Built Celery/Redis background workers for email delivery, cleanup jobs, and audit logging.
- Designed a Dockerized Nginx/FastAPI/PostgreSQL architecture with comprehensive pytest coverage.

CortexRAG · [In Development](#)

Stack: Python · FastAPI · PostgreSQL · pgvector · Hugging Face Embeddings · LangChain · Redis · Docker

- Building a RAG-based document intelligence platform (analogous to Perplexity AI / OpenAI RAG workflows) with a full ingestion pipeline: PDF upload → text extraction → intelligent chunking → vector embedding generation → pgvector cosine similarity storage → semantic retrieval → LLM-grounded response generation
- Implementing citation enforcement and confidence scoring layers to mitigate LLM hallucinations, combined with context window management via chunk re-ranking and retrieval pipelines for long-document handling and multi-hop reasoning
- Designing async document ingestion with batched Hugging Face embedding generation, Redis query caching, and streaming LLM response delivery for sub-second perceived latency at scale

RESEARCH

LLM Fine-Tuning for Automated Debate Judging

Submitted to ACL — Association for Computational Linguistics (· Annual Conference · Top-Tier NLP Venue)

- Fine-tuned 12 lightweight LLMs as automated debate judges, achieving multi-dimensional evaluation performance closely approaching human-level accuracy across argumentation quality benchmarks — directly informing the Oratoria AI scoring engine
- Demonstrated that fine-tuned lightweight models operate at approximately 1,000× lower energy consumption than commercial-scale LLMs (GPT-4/Claude) while maintaining competitive debate judging quality, with implications for sustainable AI deployment
- Conducted comparative analysis of model architectures and fine-tuning strategies for automated debate assessment.

PROFESSIONAL EXPERIENCE

Daadras Foundation — *Technology & Strategy Lead* — *Project Salaam*

Sep 2025 – Mar 2026

- Led technology adoption and operational digitization for an educational program serving 50+ active students, improving coordination, tracking, and reporting processes
- Designed data collection and performance-monitoring workflows for program evaluation.
- Coordinated between academic, operational, and technical stakeholders to align program objectives with scalable technology-driven solutions
- Developed reporting frameworks that improved visibility and collaboration.
- Evaluated and implemented digital tools to streamline administrative operations, student engagement tracking, and communication workflows

Daadras Foundation — *IT Instructor & Technical Support Associate*

Dec 2024 – Feb 2025

- Delivered structured instruction in computer literacy, programming fundamentals, and problem-solving methodologies to learners with diverse educational backgrounds
- Developed learning materials, assessment frameworks, and practical exercises to improve technical comprehension and analytical thinking skills
- Provided technical support, troubleshooting, and systems assistance across organizational software and operational infrastructure
- Assisted in maintaining and improving internal data management processes, database records, and workflow automation activities

Freelance — *Technical Tutor & Programming Mentor*

Sep 2020 – Present

- Delivered one-on-one and small-group mentorship in Python programming, computer science fundamentals, mathematics, and logical problem solving to O-Level, A-Level, and university-level students
- Designed customized learning pathways and assessments.
- Guided students in software development, algorithms, debugging, and problem solving.
- Built a track record of translating complex technical topics into clear, actionable learning experiences for audiences with varying technical backgrounds

EDUCATION

Bachelor of Computer Science (Honours)

Forman Christian College (A Chartered University) | Lahore, Pakistan

Graduated: June 2026